

Technology Stack

Date	1 July 2026
Team ID	
Project Name	AI-Powered Student Mental Wellness Platform
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	React.js, HTML5, CSS3, JavaScript, Bootstrap	React provides reusable UI components and a responsive, interactive user interface. HTML, CSS, and Bootstrap ensure a clean, mobile-friendly design.

2	Backend / Server-Side <i>(e.g., Node.js, Python/Django, Java)</i>	Node.js with Express.js	Node.js provides fast, scalable server-side processing, while Express.js simplifies API development and request handling.
3	Database / Data Storage <i>(e.g., MySQL, MongoDB, PostgreSQL)</i>	MongoDB	MongoDB efficiently stores user profiles, mood logs, chatbot history, and assessment results using flexible document-based storage.
4	Cloud / Hosting / Deployment <i>(e.g., AWS, Azure, Heroku, Docker)</i>	Render (or Vercel for frontend) and MongoDB Atlas	Render/Vercel provide easy deployment and automatic updates. MongoDB Atlas offers secure cloud database hosting with scalability.

5	Version Control & CI/CD	Git, GitHub, GitHub Actions	Git enables version control, GitHub supports team collaboration, and GitHub Actions automates testing and deployment.
	<i>(e.g., GitHub, GitLab, Jenkins)</i>		
6	Third-Party APIs / Other Tools	Google Gemini API, Firebase Authentication, Chart.js	Gemini API powers the AI mental health chatbot. Firebase Authentication provides secure user login. Chart.js visualizes mood trends and wellness analytics through interactive graphs.
	<i>(e.g., Stripe, Twilio, Google Maps)</i>		